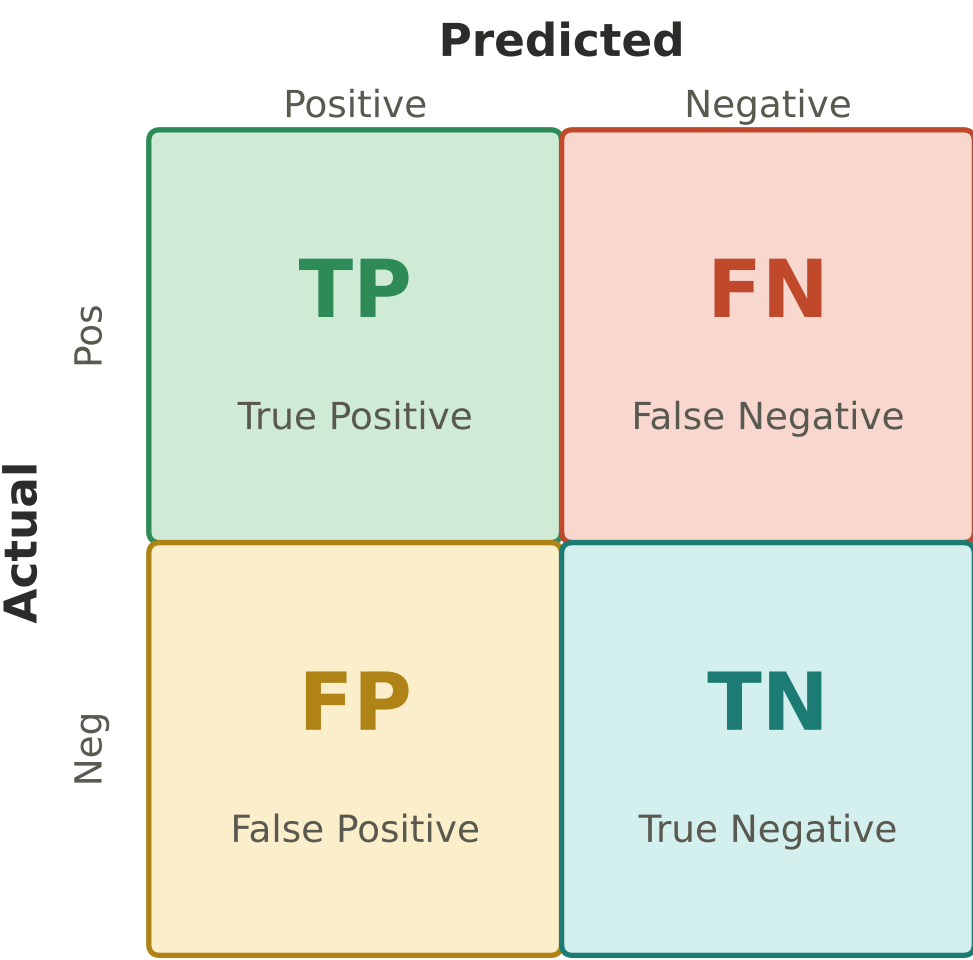


# Classification: judging a model, and four ways to draw the boundary

Turn scores into labels, score the labels across every threshold, then compare classifiers by the boundary each one draws.

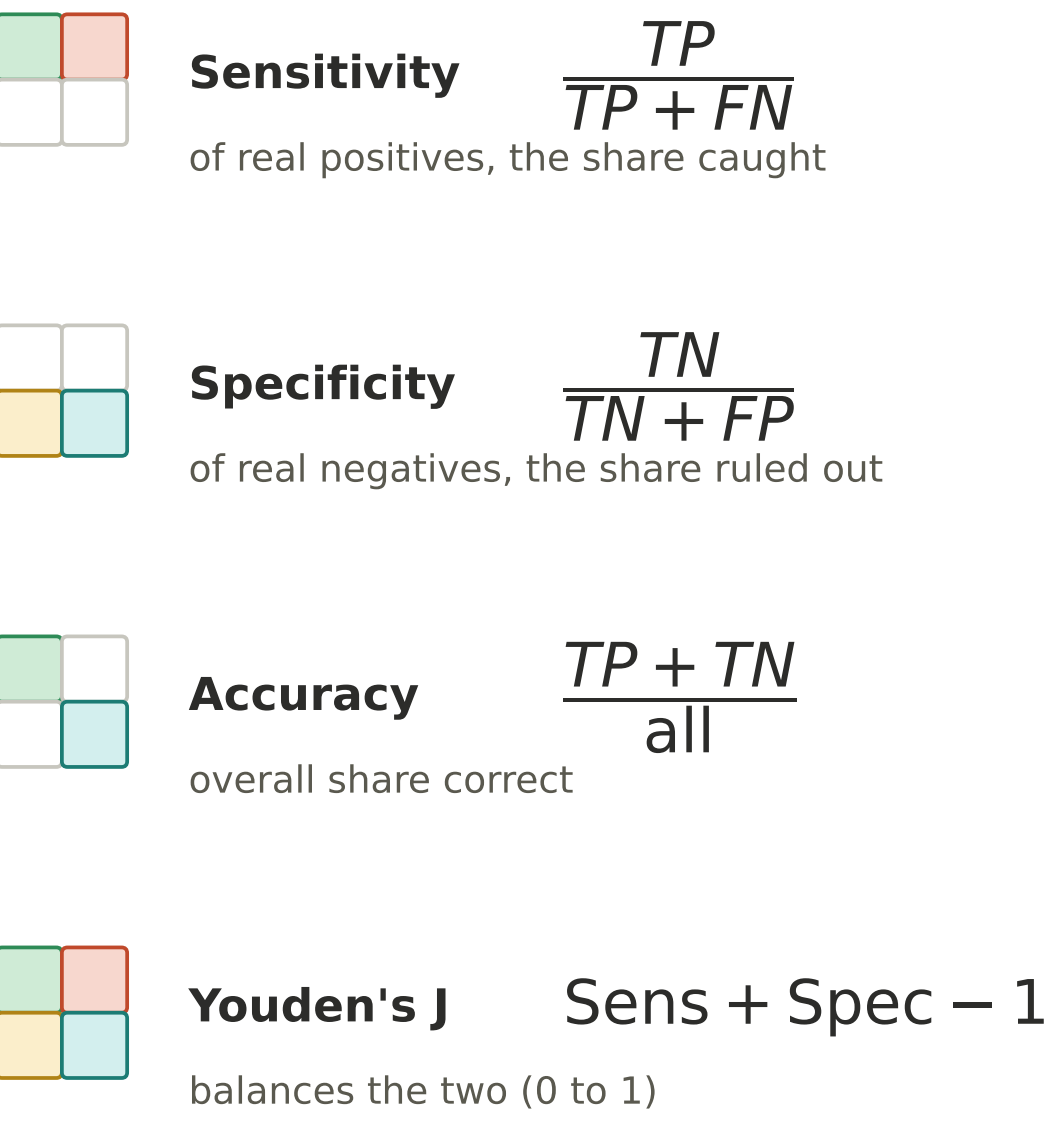
## A Judging a classifier — use every threshold, not one cutoff

### The confusion matrix

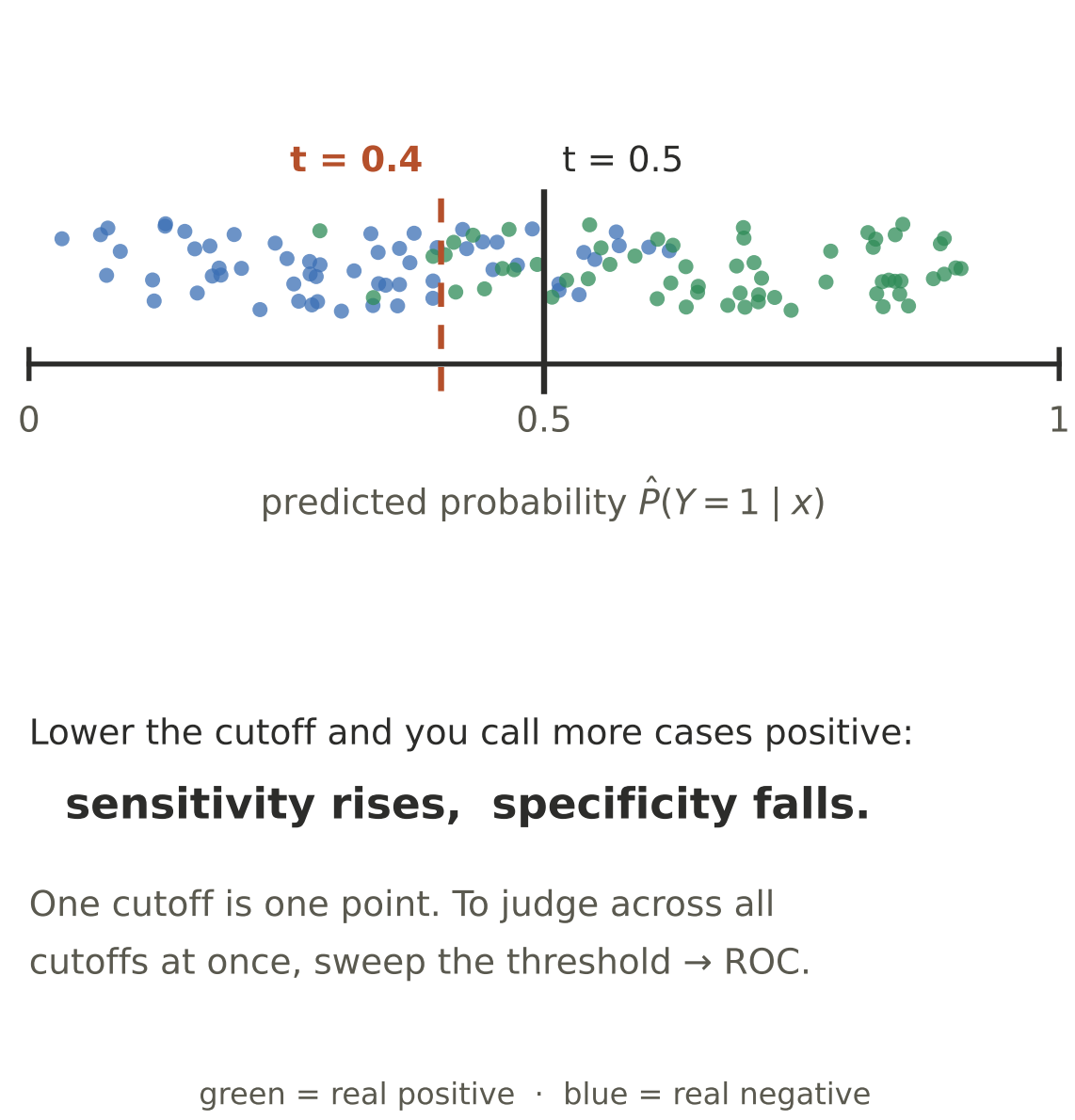


Threshold rule: classify  $\hat{Y} = 1$  when  $\hat{P}(Y = 1 | x) > t$ .

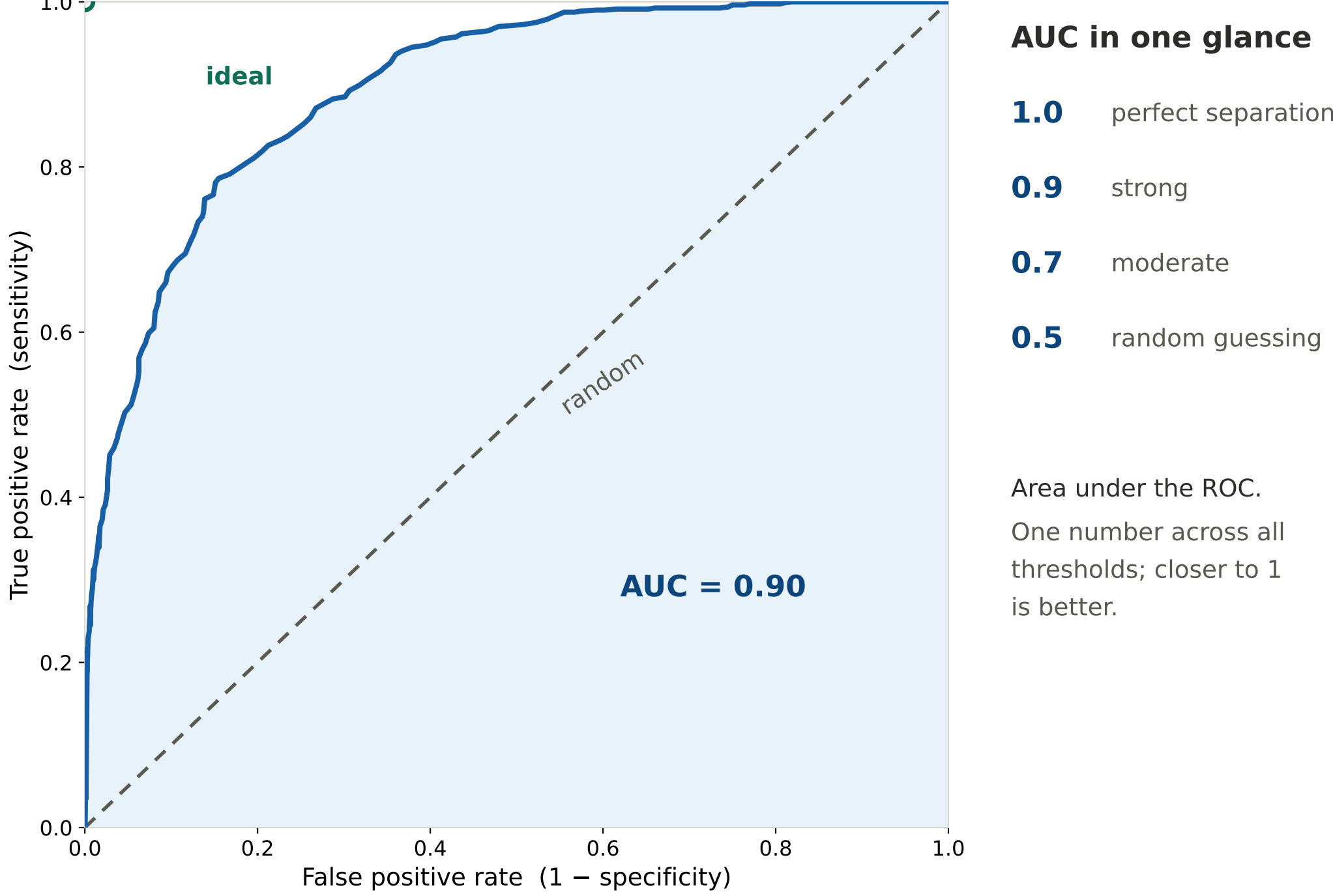
### Metrics from the counts



### Move the threshold



### ROC curve and AUC



## B The classifiers — they differ only in the covariance assumption (k-NN drops the model)

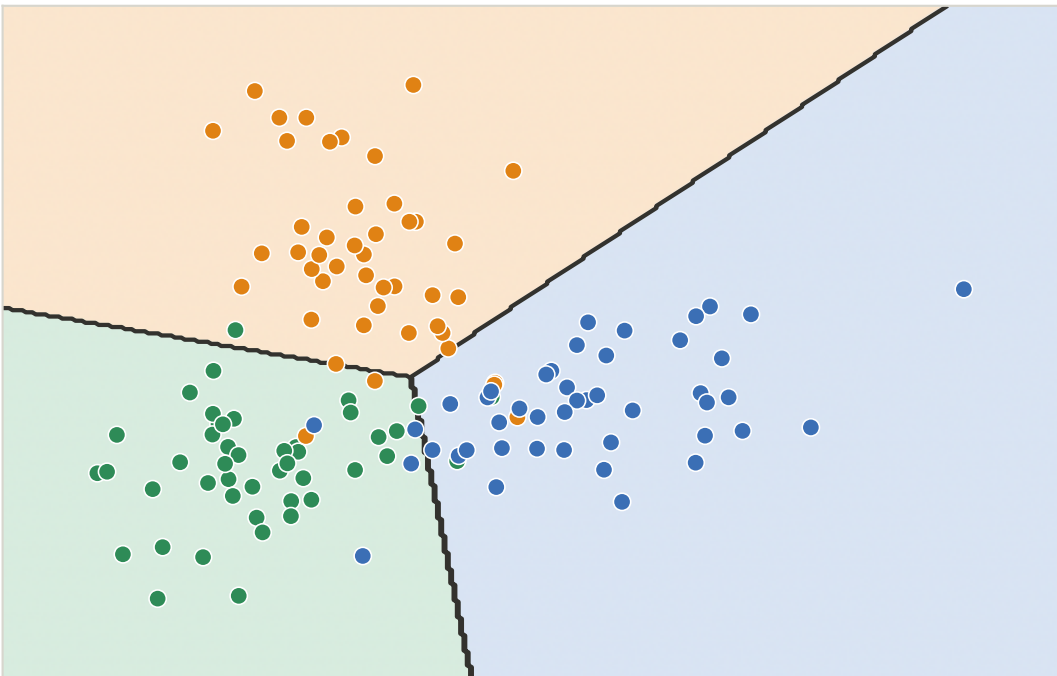
### Generative recipe (LDA, QDA, naive Bayes)

1. model each class density  $f_k(x) = \mathcal{P}(x | Y = k)$  2. flip with Bayes  $\mathcal{P}(Y = k | x) \propto \pi_k f_k(x)$  3. predict  $\hat{k} = \arg \max_k \pi_k f_k(x)$

k-NN is not generative: it fits no density and uses a majority vote of nearby points — shown at right as the non-parametric contrast.

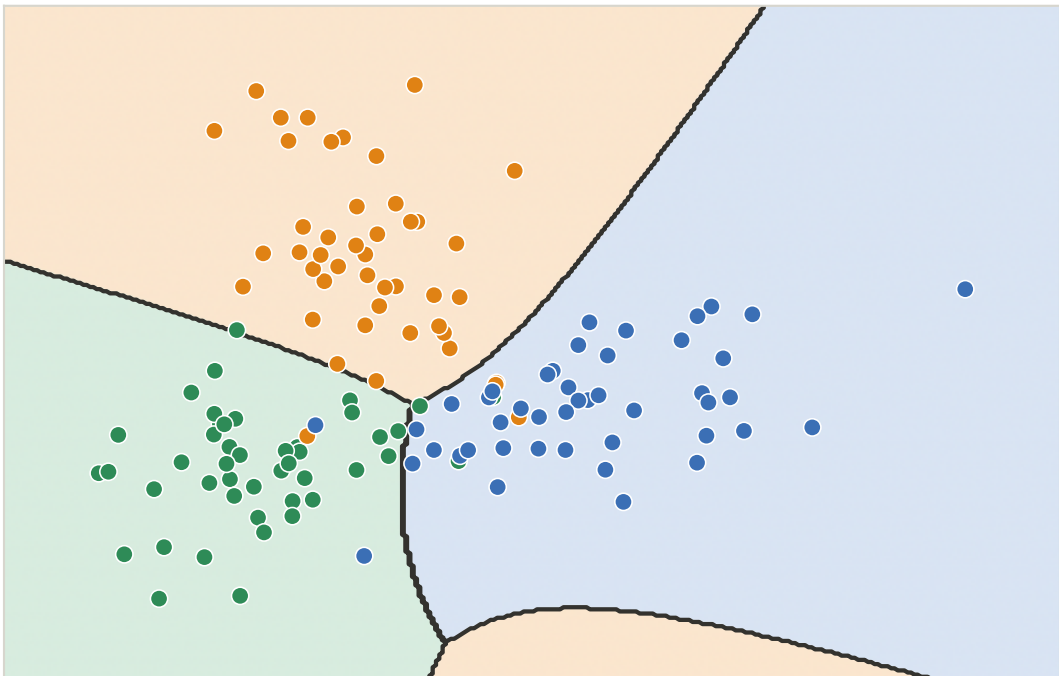
### Linear discriminant

LDA · straight lines



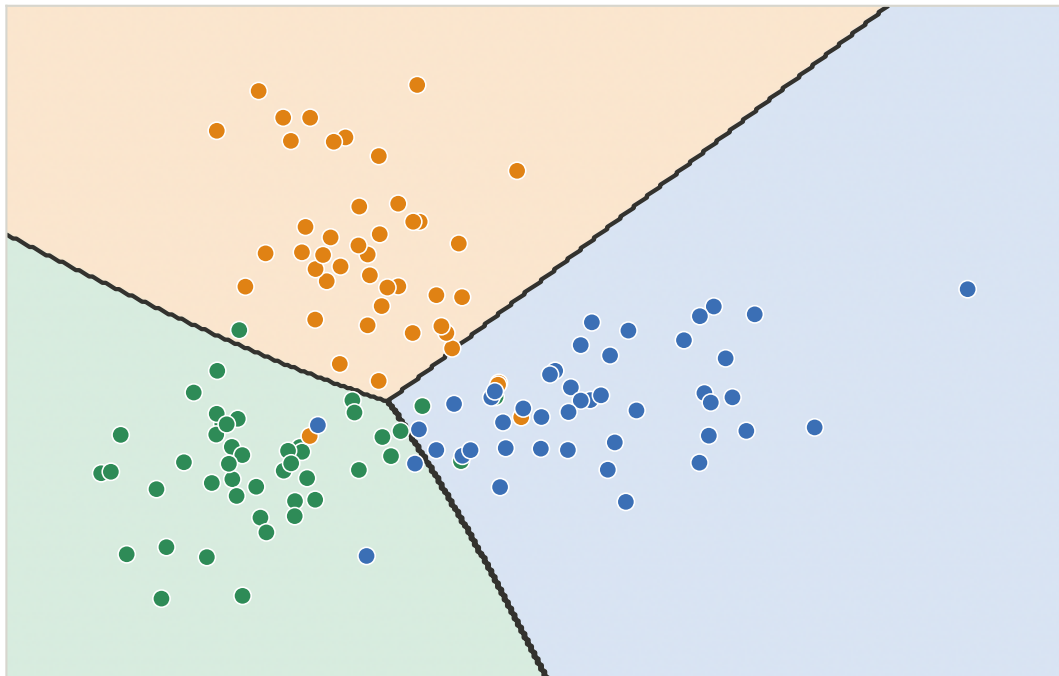
### Quadratic discriminant

QDA · curved (quadratic)



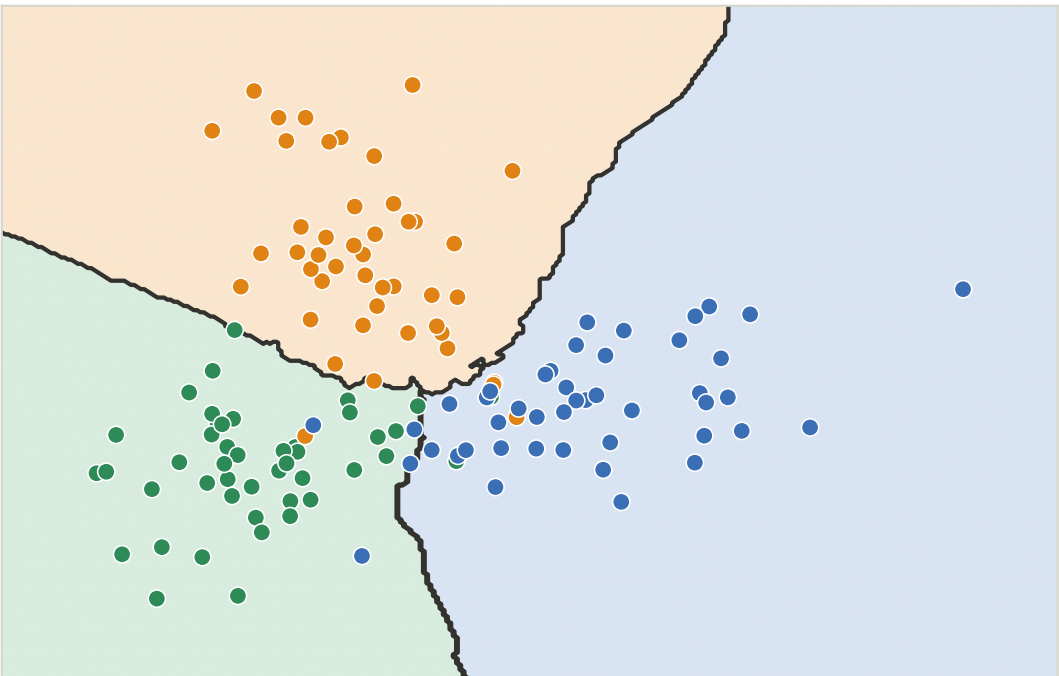
### Naive Bayes

NB · axis-aligned



### k-nearest neighbors

k-NN · jagged, data-driven



**assumption:** shared covariance  $\Sigma$

**boundary:** straight lines

**per class:** means  $\mu_k$  + one shared  $\Sigma$

**use when:** Classes  $\sim$  Gaussian with similar spread. Stable when  $n$  is small.

**assumption:** one covariance per class  $\Sigma_k$

**boundary:** curved (quadratic)

**per class:**  $\mu_k$  + full  $\Sigma_k$  each class

**use when:** Gaussian classes with different spread/shape. Needs more data.

**assumption:** predictors independent in class

**boundary:** axis-aligned

**per class:**  $\mu_k$  + variances (diagonal  $\Sigma_k$ )

**use when:** Many predictors, little data. Takes mixed-type inputs that LDA/QDA cannot.

**assumption:** none — non-parametric

**boundary:** jagged, data-driven

**per class:** just  $K$  (stores all data)

**use when:** Complex non-linear border, enough data, small  $p$ .  $K$  by cross-validation.

### The big picture

Judge with the confusion matrix, then the ROC curve and AUC to see performance across every threshold — not at one cutoff.

LDA, QDA, and naive Bayes are the same generative recipe under looser-to-tighter covariance assumptions; k-NN keeps no model and votes with neighbors. Flexibility trades against stable estimates.

green / orange / blue = the three classes